POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

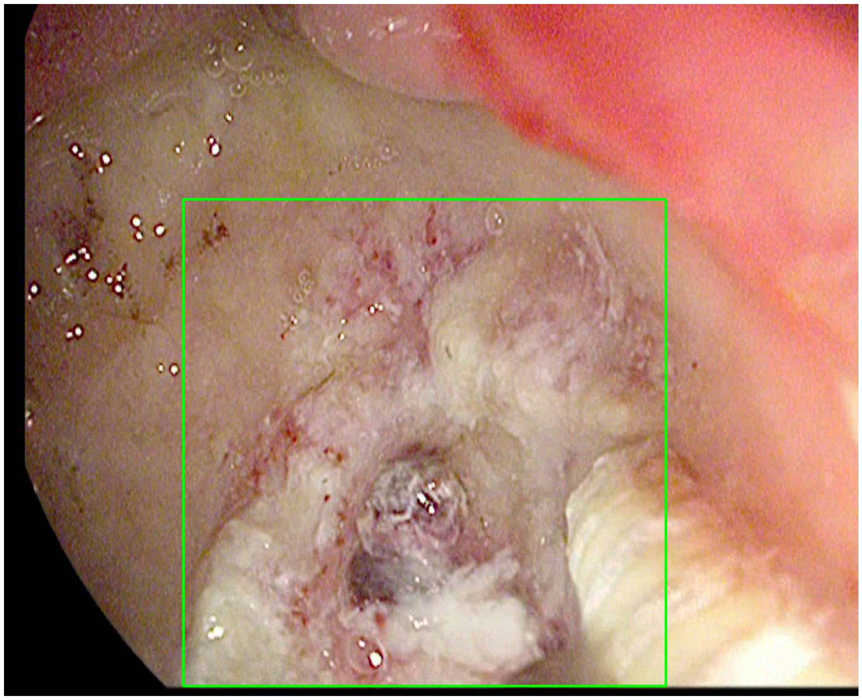
Video ID:	5c812c3f-33e6-4e1a-b708-637e76ab9244
Total Frames:	1701
Frame Rate:	25.0 fps (assumed)
Duration:	68.0 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	171	✓
RT-DETR Detections	943	✓
MedSAM2 Detections	968	✓
Consensus (3-Model Agreement)	7	✓
Risk Classification	**Count**	**Percentage**
HIGH RISK	0	0.0%
MEDIUM RISK	0	0.0%
LOW RISK	1	14.3%
TOTAL ANALYZED	7	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — LIFTED_POLYP [LOW_RISK] (Tracked across 148 frames)



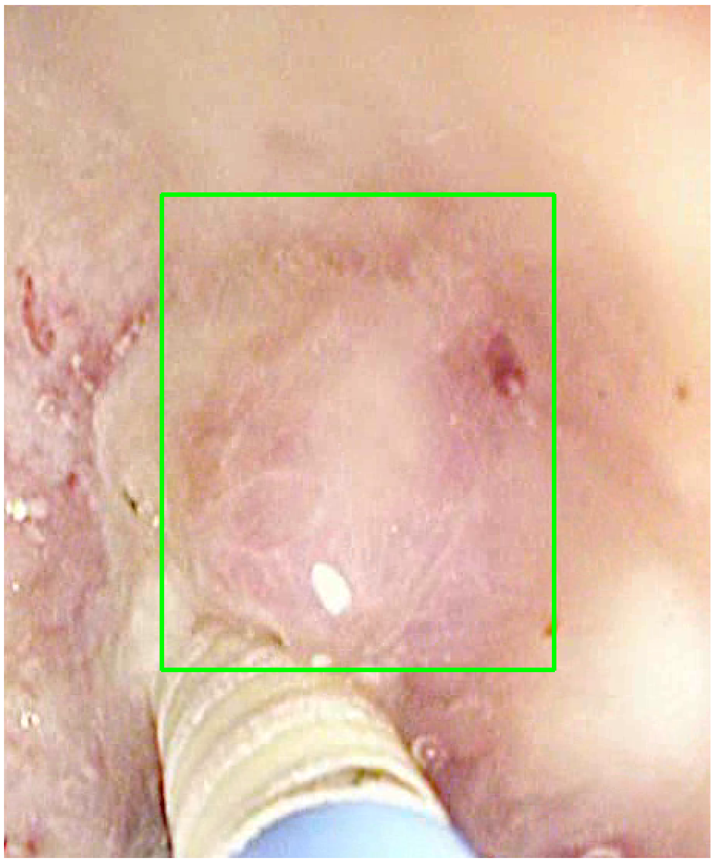
Feature	Value	Interpretation
Frame Index	677	Frame 677 of 1701
Detection Method	Detector Consensus (YOLO + RT-DETR agreement) (MedSAM2 unavailable for this video)	MedSAM2 Unavailable
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 59.4%
Redness Score	0.056	Color-based vascularization
Relative Radius	25.4%	Polyp size as % of frame diagonal
Texture Score	0.047	Surface morphology
Vessel Visibility	0.015	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output

Clinical Class	NORMAL_MUCOSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	61.5%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #2 — LIFTED_POLYP [LOW_RISK] (Tracked across 15 frames)



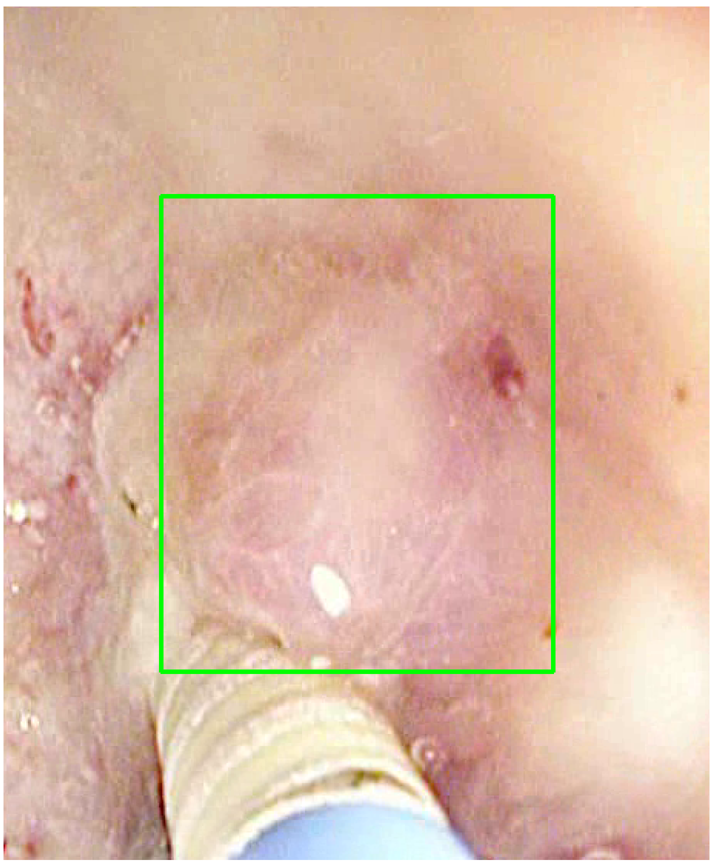
Feature	Value	Interpretation
Frame Index	568	Frame 568 of 1701
Detection Method	Detector Consensus (YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)	(MedSAM2 Unavailable)
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 60.7%
Redness Score	0.070	Color-based vascularization
Relative Radius	13.5%	Polyp size as % of frame diagonal

Texture Score	0.071	Surface morphology
Vessel Visibility	0.008	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUCOSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	66.7%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #3 — LIFTED_POLYP [LOW_RISK] (Tracked across 15 frames)



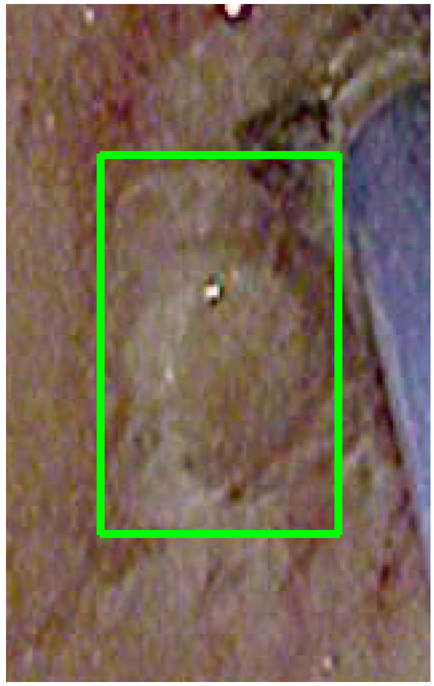
Feature	Value	Interpretation
Frame Index	568	Frame 568 of 1701

Detection Method	Detector Consensus (YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)	(MedSAM2 Unavailable)
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 60.7%
Redness Score	0.070	Color-based vascularization
Relative Radius	13.5%	Polyp size as % of frame diagonal
Texture Score	0.071	Surface morphology
Vessel Visibility	0.008	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUCOSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	58.2%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #4 — COLITIS [MEDIUM_RISK] (Tracked across 76 frames)

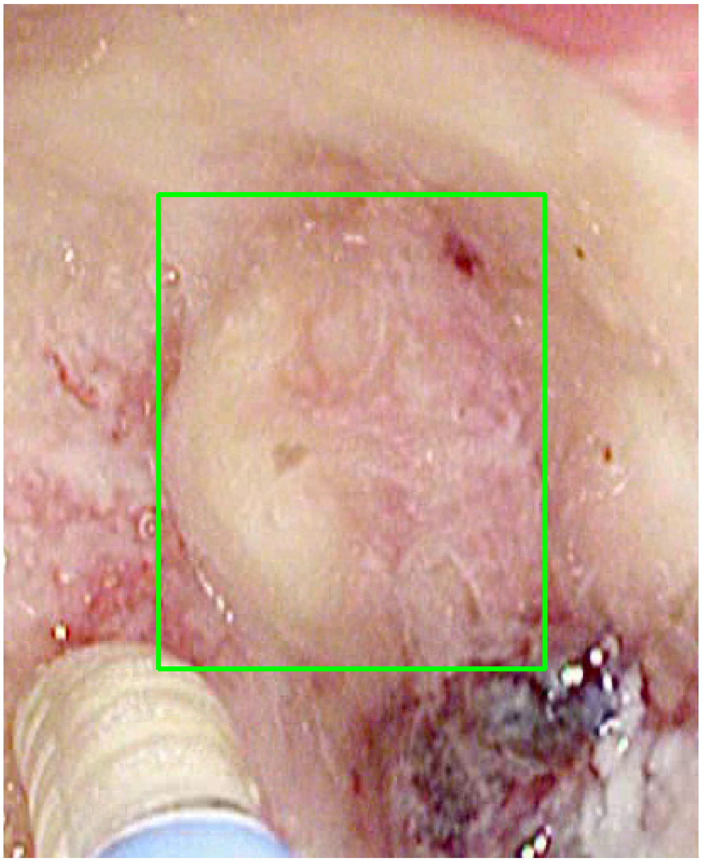


Feature	Value	Interpretation
Frame Index	1238	Frame 1238 of 1701
Detection Method	Detector Consensus (YOLO + RT-DETR agreement) (MedSAM2 unavailable for this video)	MedSAM2 Unavailable
POLYP TYPE	**COLITIS**	Type confidence: 77.6%
Redness Score	0.181	Color-based vascularization
Relative Radius	6.4%	Polyp size as % of frame diagonal
Texture Score	0.072	Surface morphology
Vessel Visibility	0.153	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	61.2%	YOLO / RT-DETR / track confidence

Type Assessment: Inflammatory mucosal pattern — compatible with IBD or infectious colitis. Clinical correlation required.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #5 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 49 frames)



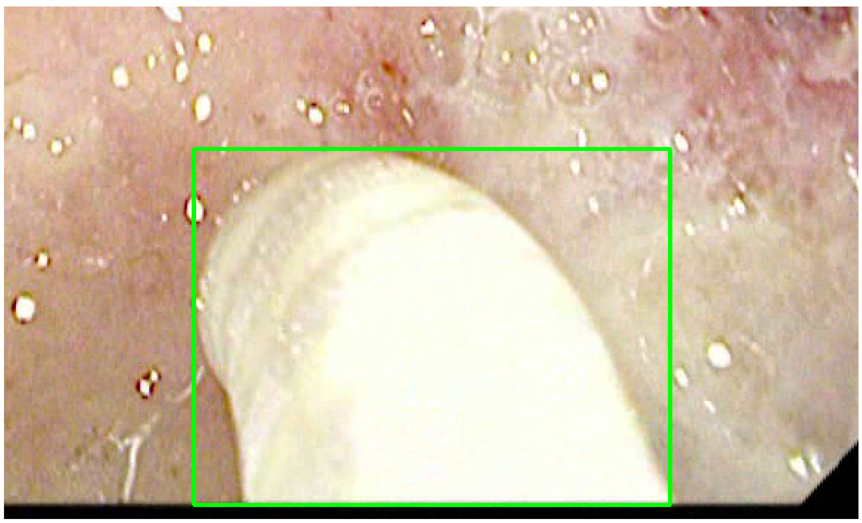
Feature	Value	Interpretation
Frame Index	396	Frame 396 of 1701
Detection Method	Detector Consensus (MedSAM2 Unavailable)	YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)
POLYP TYPE	**FLAT_POLYP**	Type confidence: 66.7%
Redness Score	0.098	Color-based vascularization
Relative Radius	23.0%	Polyp size as % of frame diagonal
Texture Score	0.075	Surface morphology
Vessel Visibility	0.060	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification

Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	NORMAL_MUCOSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	60.2%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #6 — LIFTED_POLYP [LOW_RISK] (Tracked across 41 frames)



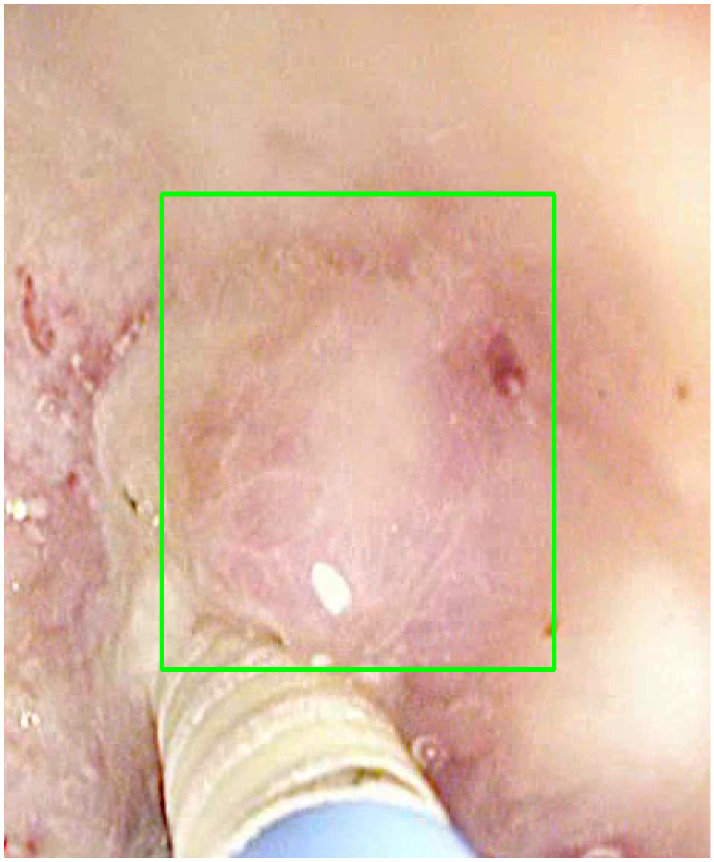
Feature	Value	Interpretation
Frame Index	296	Frame 296 of 1701
Detection Method	Detector Consensus (MedSAM2 Unavailable)	YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 54.6%
Redness Score	0.028	Color-based vascularization
Relative Radius	22.0%	Polyp size as % of frame diagonal
Texture Score	0.044	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification

Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUCOSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	88.4%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #7 — LIFTED_POLYP [LOW_RISK] (Tracked across 15 frames)



Feature	Value	Interpretation
Frame Index	568	Frame 568 of 1701
Detection Method	Detector Consensus (YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)	MedSAM2 Unavailable
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 60.7%
Redness Score	0.070	Color-based vascularization
Relative Radius	13.5%	Polyp size as % of frame diagonal
Texture Score	0.071	Surface morphology
Vessel Visibility	0.008	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output

Clinical Class	NORMAL_MUCOSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	60.4%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

CLINICAL IMPRESSION

Summary:
A total of 7 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:
• HIGH RISK polyps: 0 (0.0%)
• MEDIUM RISK polyps: 0 (0.0%)
• LOW RISK polyps: 1 (14.3%)

Recommendation:
All detected polyps are classified as LOW RISK. Standard surveillance protocol is recommended.
Technical Details:
• Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
• Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
• Classification: Cluster-specific Decision Tree experts
• Confidence: Based on deep learning + medical heuristics

Disclaimer: *This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.*

AI-Based Polyp Detection System
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